

OB'YACHEVO, Russian Federation

WMO: 229960

Lat: 60.3287N

Lon: 49.6357E

Elev: 152

StdP: 99.51

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-32.2	-28.6	-35.1	0.1	-32.1	-31.5	0.2	-28.6	7.4	-7.4	6.7	-8.2	1.1	10	0.616

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.8	29.0	20.1	27.0	19.0	24.9	17.8	21.2	26.9	20.0	25.3	19.0	23.5	2.8	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.2	14.2	23.6	18.2	13.3	22.4	17.2	12.5	21.5	62.0	26.8	58.0	25.3	54.5	23.7	27.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	7.1	6.3	5.6	DB	-33.4	31.7	3.6	1.7	-36.0	32.9	-38.1	33.9	-40.1	34.8	-42.8	36.1
(5)				WB	-33.7	23.1	3.7	2.1	-36.4	24.6	-38.5	25.8	-40.6	27.0	-43.3	28.5

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	2.5	-12.9	-11.6	-4.8	2.9	9.9	15.1	18.0	14.4	9.0	2.8	-4.5	-9.8
	DBStd	11.99	8.01	7.58	5.18	4.85	5.53	4.64	4.32	4.40	4.07	4.58	6.07	8.11
	HDD10.0	3423	710	605	458	217	73	8	2	9	66	227	436	613
	HDD18.3	5882	968	838	716	462	266	117	61	137	279	481	686	871
	CDD10.0	672	0	0	0	4	69	163	248	146	37	4	0	0
	CDD18.3	89	0	0	0	0	4	21	49	15	1	0	0	0
	CDH23.3	792	0	0	0	1	45	197	429	118	1	0	0	0
	CDH26.7	199	0	0	0	0	6	42	116	34	0	0	0	0
Wind	WSAvg	2.7	2.7	2.7	2.9	2.8	2.9	2.6	2.2	2.3	2.5	2.8	2.8	2.9
Precipitation	PrecAvg	649	39	30	37	34	57	80	64	83	61	57	52	45
	PrecMax	789	65	64	116	71	142	139	179	174	114	97	108	71
	PrecMin	516	17	8	6	2	28	26	10	20	20	23	15	28
	PrecStd	80	11	15	22	16	28	32	39	39	26	23	22	12
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.7	1.8	8.9	20.8	27.6	30.5	31.4	30.7	22.3	17.2	8.0	3.6
		MCWB	0.1	0.5	3.9	12.0	16.8	20.0	20.5	20.9	16.4	11.4	7.0	3.3
	2%	DB	0.0	0.4	5.4	15.8	24.1	27.6	29.6	26.2	19.0	13.2	4.7	1.7
		MCWB	-0.4	-0.5	1.7	8.6	14.2	18.8	20.2	19.6	14.2	10.1	3.9	1.3
	5%	DB	-1.3	-0.7	3.5	12.6	21.8	25.6	27.9	23.5	16.9	10.9	3.1	0.6
		MCWB	-1.8	-1.5	0.7	6.5	13.0	17.4	19.9	17.8	13.0	8.5	2.4	0.1
	10%	DB	-2.8	-2.2	1.9	10.2	19.2	23.5	25.8	21.3	15.2	8.8	1.7	-0.4
		MCWB	-3.2	-3.1	-0.1	5.1	12.0	16.3	18.9	16.6	12.4	7.4	1.1	-0.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.3	0.8	4.4	12.4	18.6	21.9	22.9	21.9	16.9	12.3	7.0	3.4
		MCDB	0.5	1.4	8.2	20.3	23.9	28.0	27.9	28.2	20.6	15.3	8.0	3.5
	2%	WB	-0.4	-0.3	2.5	9.3	15.8	20.2	21.4	20.3	15.3	10.4	4.1	1.4
		MCDB	-0.1	0.4	4.5	14.7	21.6	25.5	27.2	25.4	18.1	12.5	4.6	1.7
	5%	WB	-1.8	-1.5	1.2	7.2	14.2	18.9	20.5	18.7	13.8	9.0	2.5	0.1
		MCDB	-1.3	-0.7	2.9	11.5	19.0	24.0	26.1	22.1	15.7	10.7	2.9	0.4
	10%	WB	-3.2	-3.0	0.1	5.7	12.7	17.5	19.5	17.4	12.6	7.5	1.2	-0.9
		MCDB	-2.7	-2.3	1.6	9.5	18.4	21.8	24.5	20.5	14.8	8.7	1.7	-0.4
Mean Daily Temperature Range	5% DB	MDBR	5.7	6.7	7.6	8.6	9.9	9.7	9.8	8.5	6.8	4.4	3.8	4.9
		MCDBR	5.1	5.1	8.5	12.7	13.9	12.5	12.6	11.4	10.1	7.9	4.6	3.1
		MCWBR	5.2	4.7	5.8	7.2	6.8	5.5	5.1	5.3	5.7	5.2	4.4	3.2
	5% WB	MCDBR	5.0	4.9	6.4	11.0	11.9	10.9	11.2	9.9	8.4	6.6	4.2	3.1
		MCWBR	5.1	4.7	4.7	6.8	6.8	5.5	5.5	5.2	5.3	4.8	4.2	3.2
Clear-Sky Solar Irradiance	taub	0.264	0.302	0.343	0.375	0.363	0.360	0.393	0.384	0.349	0.324	0.283	0.242	
	taud	2.013	2.053	2.030	2.147	2.415	2.455	2.366	2.372	2.448	2.363	2.120	1.962	
	Ebn at Noon	461	661	756	813	863	871	827	803	774	659	442	324	
	Edh at Noon	46	81	115	124	103	101	108	99	78	61	42	31	
All-Sky Solar Radiation	RadAvg	0.33	1.03	2.23	3.71	4.89	5.19	5.25	3.72	2.12	0.88	0.30	0.17	
	RadStd	0.05	0.26	0.36	0.49	0.50	0.46	0.54	0.40	0.29	0.11	0.06	0.03	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (5 neighbors)	+0.60	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page